Prodigy Anesthesia Quick Review

Carcinoid Syndrome

What is carcinoid syndrome and what are its most common symptoms?	About 1 in 5 patients with a carcinoid tumor will experience carcinoid syndrome which results from the massive release of hormones such as serotonin. The two most common symptoms are flushing and diarrhea. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297.*
What are the three most common locations for a carcinoid tumor?	The bronchus, jejuno-ileum, and the colon. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 289.*
What types of substances can be released by a carcinoid tumor?	Carcinoid tumors most notably release serotonin and prostaglandins, but also release a wide range of peptides including gastrin, insulin, substance P, glucagon, growth hormone, and corticotropin. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297-298.*
What diagnostic test is usually indicative of carcinoid syndrome?	Carcinoid syndrome can be confirmed by detection of urinary 5-hydroindoleacetic acid. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 743.*
Not all patients with a carcinoid tumor experience side effects from the hormones released from the tumor. Why not?	Usually, hepatic first-pass metabolism eliminates the hormones from the circulatory system before they can exert any deleterious effects. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 798.*
Carcinoid syndrome often causes right-sided cardiac lesions, but typically spares the left-side unless an intracardiac shunt is present. What accounts for this?	Elevated serotonin levels can result in right-sided heart failure. The lungs metabolize serotonin, thus sparing the left side of the heart. The presence of a right-to-left shunt, however, would expose the left side of the heart to the serotonin and cause left-sided heart symptoms. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297-298.*
In which tissues does serotonin act as a vasodilator?	Serotonin acts as a vasoconstrictor in all tissues except the heart and skeletal muscle, where it acts as a vasodilator. The renal and pulmonary systems are the most sensitive to the vasoconstrictive effects of serotonin. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 743.*

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What are the cardiac manifestations of carcinoid syndrome?	Cardiac problems associated with carcinoid syndrome are usually due to fibrosis of the endocardium. Pulmonic stenosis and tricuspid regurgitation are often present. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297-298.*
How can carcinoid syndrome mimic an allergic reaction?	Some carcinoid tumors can release histamine and kallikrein which can result in bronchospasm, tachycardia, and hypotension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297-298.*
What medications relieve the diarrhea associated with carcinoid syndrome?	Serotonin antagonists such 5-HT3 antagonists relieve the diarrhea in most patients with carcinoid syndrome. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 297-298.*
What anesthesia related events should be avoided in the patient with carcinoid syndrome?	Sympathomimetics such as ephedrine and histamine-releasing drugs such as morphine, succinylcholine, thiopental, and atracurium should be avoided if possible as they can trigger the release of hormones from the tumor as well. Hypotension should also be avoided in patients with carcinoid tumors as it can be a trigger for the release of hormones from the tumor. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 798.*

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